MODULE DESCRIPTION FORM

نموذج وصف المادة الدراسية

Module Information معلومات المادة الدراسية			
Module Title	**Mobile Application Programming**	**Module Delivery**	
Module Type	**Core**	☒ Theory ☒ Lecture ☒ Lab ☒ Tutorial ☒ Practical ☐ Seminar	
Module Code	**COMP345**		
ECTS Credits	6		
SWL (hr/sem)	**150**		

Module Level	3	**Semester of Delivery**	6
Administering Department	CS	**College**	CS and Mathematics
Module Leader	Dr. Munqath Hamid Alattar	**e-mail**	munqith.alattar@uokufa.edu.iq
Module Leader's Acad. Title	Lecturer	**Module Leader's Qualification**	Ph.D.
Module Tutor	Dr. Munqath Hamid Alattar Abbass Fadhil Wahab	**e-mail**	munqith.alattar@uokufa.edu.iq abbassf.wahab@uokufa.edu.iq
Peer Reviewer Name	Name	**e-mail**	E-mail
Scientific Committee Approval Date	01/06/2023	**Version Number**	1.0

Relation with other Modules العلاقة مع المواد الدراسية الأخرى			
Prerequisite module	Programming Fundamentals	**Semester**	1
Co-requisites module	Object-Oriented Programming	**Semester**	3
Co-requisites module	Data Structure	**Semester**	3
Co-requisites module	Website Design	**Semester**	5

Module Aims, Learning Outcomes and Indicative Contents أهداف المادة الدراسية ونتائج التعلم والمحتويات الإرشادية	
Module Objectives أهداف المادة الدراسية	Course module objectives: Mobile applications have dramatically penetrated in numerous fields. The knowledge of developing an effective mobile app is vital to businesses, as almost everyone uses smartphones which in turn will increase accessibility. This module aims ● To highlight the market's high demand to employ mobile applications in business. ● To use those advanced techniques and methodologies in practical mobile developments. ● a comprehensive look at the Java programming language ● To explain the Mobile Development Lifecycle ● To use Android studio as a mobile app development IDE.
Module Learning Outcomes مخرجات التعلم للمادة الدراسية	**1. Knowledge and understanding** ● Understand the concepts of Mobile application development ● Getting familiar with the commonly used mobile application development environments and operating systems ● Exploring the available techniques, Tools and libraries. ● Using Android Studio IDE to develop mobile applications (designing UIs and working with activities and data) 2. **Cognitive skills (thinking and analysis).** ● Be able to develop basic mobile applications using Android Studio Practical skills. ● Main applications to implement (Static design, Calculator, passing data between activities, listings and details)
Indicative Contents المحتويات الإرشادية	This course tends to give the student the concepts of Mobile application development, the environments, operating systems, available techniques, and other Tools and libraries. The mobile system includes a mobile device, operating system, wireless network, mobile app, and app platform. Android Studio IDE and the Object Oriented Programming Language (Java) are used to implement mobile applications.

Learning and Teaching Strategies استراتيجيات التعلم والتعليم	
Strategies	The main strategy that will be adopted in delivering this module is to encourage students' participation in the exercises, while at the same time refining and expanding their critical thinking skills. This will be achieved through classes, interactive tutorials, practical, lab hours and by considering types of some small projects that are interesting to the students.

Student Workload (SWL) الحمل الدراسي للطالب محسوب لـ ١٥ اسبوعا			
Structured SWL (h/sem) الحمل الدراسي المنتظم للطالب خلال الفصل	80	**Structured SWL (h/w)** الحمل الدراسي المنتظم للطالب أسبوعيا	2
Unstructured SWL (h/sem) الحمل الدراسي غير المنتظم للطالب خلال الفصل	45	**Unstructured SWL (h/w)** الحمل الدراسي غير المنتظم للطالب أسبوعيا	2
Total SWL (h/sem) الحمل الدراسي الكلي للطالب خلال الفصل	**125**		

Module Evaluation تقييم المادة الدراسية					
		Time/Number	**Weight (Marks)**	**Week Due**	**Relevant Learning Outcome**
Formative assessment	**Quizzes**	2	10% (10)	5 and 10	All
	Assignments	4	10% (10)	4,8 and 12	All
	Projects / Lab.	1	10% (10)	6 to 14	All
	Report	1	10% (10)	2	All
Summative assessment	**Midterm Exam**	2hr	10% (10)	7 and 13	All
	Final Exam	3hr	50% (50)	16	All
Total assessment			100% (100 Marks)		

Delivery Plan (Weekly Syllabus) المنهاج الاسبوعي النظري	
	Material Covered
Week 1	Introduction to mobile application development,
Week 2	Mobile Operating Systems, OS distribution. Android and iOS.
Week 3	Available mobile app Development techniques: Android Studio, xCode Flutter, React Native, Xamarin, and Web Apps.
Week 4	Essential Concepts and Terms: SDK, Virtual machines, Emulator, IDEs, XML, and API.
Week 5	Usability and User Interaction Design. Mobile Navigation and Interface Design: Activities, constraints, and layouts. (The concepts)
Week 6	Introduction to Android Studio, First application (Hello World), debug the application and create an APK.
Week 7	First Exam
Week 8	Designing Mobile Application activities using XML
Week 9	Layout (Linear, Relative and Constraints)

Week 10	Working with View Data (getting and setting data from Views such as TextView, EditText, Buttons, ...)
Week 11	Working with multiple activities, sending data between Activities
Week 12	Listings concepts (ArrayList and ListView)
Week 13	Second Exam
Week 14	Revision and project discussion
Week 15	Final Exam

Delivery Plan (Weekly Lab. Syllabus)

المنهاج الاسبوعي للمختبر

	Material Covered
Week 1	Lab 1: Java Programming revision
Week 2	Lab 2: Reports and statistics about Mobile Development Techniques
Week 3	Lab 3: Working with XML (HW: Write an XML code of a Mobile UI design)
Week 4	Lab 4: Java Programming fundamentals for Android development
Week 5	Lab 5: Installing and Getting familiar with Android Studio
Week 6+7	Lab 6 and 7: Hello World Application and create an APK
Week 8	Lab 8: create applications with simple UI
Week 9	Lab 9: UI design using Costraint Layouts (Using the constraints to fix views positions)
Week 10	Lab 10: the Calculator application Adding and Subtracting numbers of two EditTexts. (HW: Static Login Application).
Week 11	Lab 11: Multiple activities Calculator application (HW: Multiple activities Login Application)
Week 12	Lab 12: Listing Application
Week 13	Lab 13: Menu and ActionBar Application (HW: Application with multiple activities listing interaction and ActionBar with menu)
Week 14	Lab 14: Discussing the HW in Week 13
Week 15	Final Exam

Learning and Teaching Resources

مصادر التعلم والتدريس

	Text	Available in the Library?
Required Texts	**Mobile Applications Development with Android: Technologies and Algorithm,** Keke Gai, Maikang Qiu, and Wenyun Dai	Yes
Recommended Texts	**Java for Absolute Beginners**, Iuliana Cosmina, 2018 **Android Programming for Beginners,** John Horton	Yes
Websites	https://developer.android.com/ https://www.w3schools.com/java/ https://www.codecademy.com/learn/learn-java https://www.javatpoint.com/java-tutorial	

Grading Scheme مخطط الدرجات				
Group	**Grade**	التقدير	**Marks %**	**Definition**
Success Group (50 - 100)	**A -** Excellent	امتياز	90 - 100	Outstanding Performance
	B - Very Good	جيد جدا	80 - 89	Above average with some errors
	C - Good	جيد	70 - 79	Sound work with notable errors
	D - Satisfactory	متوسط	60 - 69	Fair but with major shortcomings
	E - Sufficient	مقبول	50 - 59	Work meets minimum criteria
Fail Group (0 – 49)	**FX –** Fail	راسب (قيد المعالجة)	(45-49)	More work required but credit awarded
	F – Fail	راسب	(0-44)	Considerable amount of work required

Note: Marks Decimal places above or below 0.5 will be rounded to the higher or lower full mark (for example a mark of 54.5 will be rounded to 55, whereas a mark of 54.4 will be rounded to 54. The University has a policy NOT to condone "near-pass fails" so the only adjustment to marks awarded by the original marker(s) will be the automatic rounding outlined above.